OS Orange Problem 2

PES VCS

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Phase 1:

Screenshot 1A:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Screenshot 1B:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ find .pes/objec
ts -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Phase 2:

Screenshot 2A:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Screenshot 2B:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ find .pes/objec
ts -type f
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ xxd .pes/object
s/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
00000000: 626c 6f62 2031 3200 6865 6c6c 6f20 776f  blob 12.hello wo
00000010: 726c 640a                                rld.
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Phase 3:

Screenshot 3A:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./pes add file1
.txt file2.txt
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.h
  untracked:  README.md
  untracked:  pes.h
  untracked:  .gitignore
  untracked:  test_objects.c
  untracked:  tree.c
  untracked:  object.c
  untracked:  index.c
  untracked:  test_objects
  untracked:  pes.c
  untracked:  commit.c
```

Screenshot 3B:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 17766931
36 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 17766931
36 6 file2.txt
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Phase 4:

Screenshot 4A:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ ./pes log
commit 84efb8aea143b321ed0cc650ee94023b00b86804d7c34364afdc13b25d7f0831
Author: Tanmay T A <PES1UG24CS492>
Date:   1776695615

    Add farewell

commit 963006942df52bfcc07b942c6e462263eba0461a5de544085fca8155612fd2de
Author: Tanmay T A <PES1UG24CS492>
Date:   1776695615

    Add world

commit fa1f2bc62fef784c807b0b5b8df135c38efb4512ab05a7609445f53d7ac65009
Author: Tanmay T A <PES1UG24CS492>
Date:   1776695614

    Initial commit

commit 610bf98da7e87e0edbc70f279c43fc7a0fd62beb2e0da44eb6c338f1890b181a
Author: Tanmay T A <PES1UG24CS492>
Date:   1776695590

    Initial commit
```

Screenshot 4B:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ find .pes -type
 f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/61/0bf98da7e87e0edbc70f279c43fc7a0fd62beb2e0da44eb6c338f1890b181a
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/84/efb8aea143b321ed0cc650ee94023b00b86804d7c34364afdc13b25d7f0831
.pes/objects/96/3006942df52bfcc07b942c6e462263eba0461a5de544085fca8155612fd2de
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/fa/1f2bc62fef784c807b0b5b8df135c38efb4512ab05a7609445f53d7ac65009
.pes/refs/heads/main
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Screenshot 4C:

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ cat .pes/refs/h
eads/main
84efb8aea143b321ed0cc650ee94023b00b86804d7c34364afdc13b25d7f0831
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Full Integration Test Screenshot

```
tanmayta@TANMAY-MINT: ~/PESU Depression/OS/PES1UG24CS492-pes-vcs



--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
97bd9ecc2707f8cb195b60f446b75537b032801007776a75ba2a763ba66c09ea

--- Object Store ---
Objects created:
10
.pes/objects/06/1bd8a4c07c36f7efa318a402f284bc78a9a5250e4c8d60242ffec2f108dff6
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/17/0e8c7266ff091ef3f3b232c7f8ed26ec112ac7a592464190d3eb0b4ca26db3
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/97/bd9ecc2707f8cb195b60f446b75537b032801007776a75ba2a763ba66c09ea
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
tanmayta@TANMAY-MINT:~/PESU Depression/OS/PES1UG24CS492-pes-vcs$
```

Phase 5:

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

Implementing pes checkout basically means you have to update the HEAD file so it points to the new branch ref, like ref: refs/heads/main. The real headache is syncing the working directory; you have to delete files that aren't in the new branch, update the ones that changed, and add the new ones, all while making sure you don't accidentally overwrite a user's unsaved local work.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

To catch a "dirty" directory conflict, I'd just compare the mtime and size of files in the working directory against what's in the current index. If those don't match, the file is modified. Then, I'd check if that file's hash in my current index is different from the hash in the target branch's tree. If both are true, it's a conflict, and I'd have to block the checkout so the user doesn't lose their data.

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

When you're in a "Detached HEAD," it just means HEAD has a raw commit hash in it instead of a branch name. You can still make commits, but they aren't attached to any branch, so they become "orphans" the moment you switch away to something else. To get them back, you'd have to find that specific hash in the object store and manually create a new branch ref that points to it.

Phase 6:

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

To clean up the object store, I'd use a mark-and-sweep algorithm. First, I'd start at the "roots"—all the branch hashes and HEAD—and recursively follow every tree and parent pointer to "mark" every object that is still reachable. For a huge repo with 100k commits, I'd probably end up visiting a few hundred thousand objects at least, including all the trees and blobs linked to those commits. Finally, I'd just "sweep" through the .pes/objects folder and delete anything that wasn't marked.

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this?

Running GC while someone is committing is super risky because of race conditions. For example, a new commit might calculate a hash for a file and assume it's already in the store, but the GC might see that same object as "unreachable" at that exact second and delete it before the commit finishes. Real Git fixes this by having a grace period, where the GC won't touch any unreachable object unless it's at least two weeks old, giving new operations plenty of time to finish.